Star classification

Stars can be classified in many ways, but the system preferred by astronomers places the majority into seven main classes (O to M) based on their spectra—the light of various wavelengths received from them. A star's spectrum contains data relating to its color, temperature, composition, and other properties. In an attempt to see if there is any underlying pattern to the whole range of different stars, in around 1911 and 1913, Danish astronomer Ejnar Hertzsprung and American astronomer Henry Norris Russell independently plotted hundreds of stars on a scatter diagram according to their spectral class on one axis and luminosity (related to brightness) on the other. This revealed something interesting. Most stars fall into, and spend much of their lives in, a part of the diagram called the main sequence. Other parts are filled by giant stars—known to be nearing the end of their life—and by expired giant stars called white dwarfs.

▽ **The Hertzsprung–Russell diagram**
Running diagonally across the diagram is the main sequence—an array of stable stars, ranging from cool red dwarf stars to hotter, bigger, bluish stars. Other parts are occupied by stars that were once on the main sequence but later evolved into luminous giants, and by white dwarfs.